🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 5 · STANDARD 6.GR.4

Determine Surface Area of Pyramids

Lesson 5-8 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

My Notes Level 2 Standard

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can find the surface area of a pyramid by adding the base area and the lateral faces.

Language Objective: I can explain my work using the words pyramid, slant height, lateral face, and base.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Surface Area of Pyramids | Pyramid | Slant height | Lateral face | Base | Apex | Lateral area

👆 *Tap any word to see what it means and a picture.*

1. The total area of a pyramid's faces, found with SA = B + lateral area, is the ________.

2. A solid with a polygon base and triangular faces meeting at one point is a ________.

3. The height of a triangular face of a pyramid, measured along the slanted side, is the ________.

4. A triangular side face of a pyramid is a ________.

5. The polygon on the bottom of a pyramid is the ________.

6 The single top point where the triangular faces of a pyramid meet is the

________________.

7 The combined area of all the side faces, not counting the base, is the

________________.

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

How much will the glass for all 4 triangular sides cost? (The base is open — no glass needed.)?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

- [ ] **Surface Area of Pyramids** — The total area of a pyramid's base and triangular faces. Formula: SA = B + lateral area.
  Área de superficie de pirámides — El área total de la base y las caras triangulares de una pirámide. Fórmula: SA = B + área lateral.

- [ ] **Pyramid** — A solid with a flat bottom and triangle sides that meet at one point on top.
  Pirámide — Un sólido con un fondo plano y lados triangulares que se unen en un punto arriba.

- [ ] **Slant height** — The height of a side triangle, measured along its slanted face.
  Altura inclinada (apotema) — La altura de un triángulo lateral, medida por su cara inclinada.

- [ ] **Lateral face** — A triangle side of a pyramid, not the bottom.
  Cara lateral — Un lado triangular de una pirámide, no el fondo.

- [ ] **Base** — The flat bottom of a pyramid.
  Base — El fondo plano de una pirámide.

Say it first: Say your idea to a partner or quietly to yourself before you write.

Di tu idea a un compañero o en voz baja antes de escribir.

Each triangular face has area = ½ × ____ × ____ = ____ ft². Four faces: 4 × ____ = ____ ft². Cost: ____ × $12 = $____.

Di tu respuesta primero: Mi respuesta es ____ porque ____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: Surface area of a pyramid is the base area plus the areas of all the triangular lateral faces.

Evidence: Students explain a square pyramid's surface area = base area + the four triangular lateral faces, and that you add every face together.

Reasoning: This evidence connects the definition of surface area of pyramids to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **Each triangular face has area = ½ × ____ × ____ = ____ ft². Four faces: 4 × ____ = ____ ft². Cost: ____ × $12 = $____.**
- **My answer is ____.**
  Mi respuesta es ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
  Mi afirmación es ____.
- **I know because ____.**
  Lo sé porque ____.
- **So ____.**
  Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
  Mi afirmación es ____.
- **My math evidence is ____.**
  Mi evidencia matemática es ____.
- **This proves ____ because ____.**
  Esto demuestra ____ porque ____.

4. Check Your Explanation

- [ ] I answered the question.
  Respondí la pregunta.
- [ ] I used math evidence: numbers, an equation, a model, or a comparison.
  Usé evidencia matemática: números, una ecuación, un modelo o una comparación.
- [ ] I explained how the evidence proves my answer.
  Expliqué cómo la evidencia demuestra mi respuesta.
- [ ] I used at least two math words.
  Usé por lo menos dos palabras matemáticas.
- [ ] I reread complete sentences and punctuation.
  Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** Surface Area of Pyramids
2. **Notes 2:** Pyramid
3. **Notes 3:** Slant height
4. **Notes 4:** Lateral face
5. **Notes 5:** Base
6. **Notes 6:** Apex
7. **Notes 7:** Lateral area
8. **Watch for this mistake:** *A common mistake in Surface Area of Pyramids is finding the four triangular lateral faces correctly (½ × base edge × slant height, times 4) and then stopping — forgetting to add the base area at the end. This leaves the total surface area too small by exactly the base's area. Before you submit, ask: 'Did I add the base area to the four lateral faces, or did I stop after the sides?'*
9. **Try It 1:** A. 81 in² — *The base is a square, so base area = side × side = 9 × 9 = 81 in². You would still add the four triangular faces to get the full surface area.*
10. **Try It 2:**